

Topology and Manifolds

MH3600

Chapter 4: Completeness and Fixed Point Theorems

Assoc. Prof. François J. M. Gay-Balmaz

Division of Mathematical Sciences
School of Physical and Mathematical Sciences
Nanyang Technological University

Outline

- 1 Completeness and Banach spaces
- 2 Fixed Point Theorems
- 3 Application 1: Newton's Method
- 4 Application 2: Initial-Value Problems and Picard's Method
- 5 References

Outline

- 1 Completeness and Banach spaces
- 2 Fixed Point Theorems
- 3 Application 1: Newton's Method
- 4 Application 2: Initial-Value Problems and Picard's Method
- 5 References

Complete metric spaces

Definition

Let (X, d) be a metric space. A sequence (x_n) of points of X is a **Cauchy sequence** on (X, d) if for all $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that if $m, n \geq N$ then $d(x_n, x_m) < \varepsilon$.

The metric space (X, d) is **complete** if every Cauchy sequence in X converges.

By the **Triangle Inequality** for any metric, a convergent sequence is always **Cauchy** (whether the space is complete or not).

Example

It is a fundamental fact in real and complex analysis that $\mathbb{R}$ and $\mathbb{C}$ with the metrics induced by the absolute value and the modulus function, respectively, are **complete**.

Banach spaces

Definition

Let V be a vector space and let $\|\cdot\| : V \longrightarrow \mathbb{R}$ be a norm defined on it. We say that the **normed space** $(V, \|\cdot\|)$ is a **Banach space** when V is a **complete metric space** with the distance determined by the norm.

Example

The normed spaces $(\mathbb{R}^n, \|\cdot\|_p)$, $(\ell^p, \|\cdot\|_p)$ are Banach spaces for any $1 \leq p \leq \infty$.

Closed subspaces of complete spaces are complete

Theorem

Let (X, d) be a complete metric space, and let Y be a subspace of X . Then Y is complete if and only if it is closed.

Proof

We recall that we can think of Y as a metric space by using the restriction d_Y of the metric d to Y ; the restricted metric induces the subspace topology on Y (see Tutorial 4).

We assume first that Y is complete as a subspace of X , and we show that it is closed. Let y be a point in $\bar{Y}$. By the Sequence Lemma, there is a sequence $\{y_n\} \subset Y$ that converges to y in X . In particular $\{y_n\}$ is a Cauchy sequence in X and hence in Y and since Y is complete, it must converge in Y , hence $y \in Y$. This proves $\bar{Y} \subset Y$, which is therefore closed.

Proof (continued)

We now assume that Y is closed, and we show that it is complete. Let $\{y_n\}$ be a Cauchy sequence in Y . It is also a Cauchy sequence in X , and since X is complete, $\{y_n\}$ converges to a point x in X . By the sequence lemma, since $\{y_n\} \subset Y$, we have $x \in \bar{Y} = Y$, since Y is closed. Hence the Cauchy sequence converges in Y , so Y is complete. ■

Example

The sets c and c_0 of convergent and convergent real sequences with zero limit, respectively, are closed vector subspaces (prove it!) of the Banach space $(\ell^\infty, \|\cdot\|_\infty)$. The previous theorem implies that $(c, \|\cdot\|_\infty)$ and $(c_0, \|\cdot\|_\infty)$ are hence Banach spaces.

Example

The normed space $(\mathcal{B}(X, \mathbb{R}), \|\cdot\|_\infty)$ introduced in Chapter 3 is complete and hence it is a Banach space. (See [Simmons, Exercise 4]).

The Uniform Limit Theorem and the exercise given in Chapter 3, guarantee that the vector subspace $(\mathcal{C}_B(X, \mathbb{R}), \|\cdot\|_\infty)$ of bounded continuous functions is closed in $(\mathcal{B}(X, \mathbb{R}), \|\cdot\|_\infty)$ and hence by the previous theorem it is also a Banach space in its own right.

Chap. 3:

- Let $f_n : X \rightarrow Y$ be a sequence of continuous and bounded functions from the topological space X to the metric space Y . If $\{f_n\}$ converges uniformly to f , then f is continuous.
- The uniform convergence corresponds to the convergence with respect to the norm $\|\cdot\|_\infty$.

Example (Endomorphisms of $\mathbb{R}^n$)

Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$ and let $A : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a **linear map**. The norm $\|\cdot\|$ on $\mathbb{R}^n$ induces a **matrix** or **operator norm** $\|A\|$ on the space $L(\mathbb{R}^n, \mathbb{R}^n)$ of linear maps $A : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ (endomorphisms of $\mathbb{R}^n$) defined as

$$\|A\| = \max_{\|v\|=1} \|Av\| = \max_{\|v\|\neq 0} \frac{\|Av\|}{\|v\|}.$$

This norm makes the space $L(\mathbb{R}^n, \mathbb{R}^n)$ into a normed space and

$$\|Av\| \leq \|A\| \|v\|, \quad \text{for all } v \in \mathbb{R}^n.$$

It is a Banach space (all finite dimensional normed vector spaces are complete).

Example (Three different matrix norms)

Consider $\mathbb{R}^n$ with the three norms $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ that we introduced before. Write $A = (a_{ij})_{i,j \in \{1, \dots, n\}}$ a $n \times n$ matrix.

The corresponding matrix norms $\|A\|_1$, $\|A\|_2$, and $\|A\|_\infty$ are given by

- $\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| = \max_{1 \leq j \leq n} \|c_j\|_1$ (maximum column 1-norm).
- $\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| = \max_{1 \leq i \leq n} \|r_i\|_1$ (maximum row 1-norm).
- $\|A\|_2 = \sqrt{\lambda_{\max}(A^\top A)}$, that is, the **largest singular value** of A , where $\lambda_{\max}(A^\top A)$ is the maximal eigenvalue of $A^\top A$.

All the matrix norms $\|A\|$ satisfy that

$$|\lambda| \leq \rho(A) = \max\{|\lambda_1|, \dots, |\lambda_n|\} \leq \|A\|, \text{ for any eigenvalue } \lambda \in \mathbb{C} \text{ of } A.$$

The Banach space of linear operators (quick overview)

The matrix norms $\|A\|$ introduced in the previous example can be extended to rectangular matrices $A \in \mathbb{M}_{m,n}$.

Moreover, if we think of these matrices as linear maps $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the corresponding space $(L(\mathbb{R}^n, \mathbb{R}^m), \|\cdot\|)$ is a Banach space (since finite dimensional).

This construction can be extended to the space of linear maps (called operators) between arbitrary (and potentially infinite dimensional) Banach spaces.

However, in infinite dimensions, the linearity of an operator A does NOT imply neither its continuity nor its boundedness (meaning that its operator norm $\|A\|$ is finite).

Example

Consider the normed space $(C^1([-1, 1]), \|\cdot\|_\infty)$.

This is **not a Banach space** since, in general, **the uniform limit of differentiable functions is not differentiable** (a simple counterexample is the sequence $f_n(x) = \sqrt{(x - 1/2)^2 + 1/n}$, $n \in \mathbb{N}$, which converges uniformly to the non-differentiable function $f(x) = |x - 1/2|$).

Recall: a subspace of a complete space is complete if and only if it is closed.

Consider the linear map between normed spaces

$$A : (C^1([-1, 1]), \|\cdot\|_\infty) \longrightarrow (\mathbb{R}, \|\cdot\|), \quad f \longmapsto Af = f'(0). \quad (1)$$

This map is linear but not continuous. Indeed, consider the sequence $f_n(x) = \frac{\sin(n^2 x)}{n}$, for $n \geq 1$. This sequence converges uniformly to the zero function, but

$$A(f_n) = \frac{n^2 \cos(n^2 \cdot 0)}{n} = n \rightarrow \infty.$$

The norm of a linear operator

Let $(E, \|\cdot\|)$ and $(E', \|\cdot\|)$ be two linear normed spaces and let $A : E \longrightarrow E'$ be a linear map. We define the **norm** $\|A\|$ of A by

$$\|A\| = \sup\{\|A(v)\| : \|v\| \leq 1\}.$$

When $E \neq \{0\}$, this formula can clearly be written in the equivalent form

$$\|A\| = \sup\{\|A(v)\| : \|v\| = 1\}.$$

Another equivalent expression for the norm of A is:

$$\|A\| = \inf\{K : K \geq 0 \text{ and } \|A(v)\| \leq K\|v\| \text{ for all } v\}.$$

From this we see at once that $\|A(v)\| \leq \|A\|\|v\|$, for all $v \in E$. In infinite dimensions **the operator norm is NOT necessarily finite** since the unit ball is not compact when E is infinite dimensional (sup no max!).

Example $\|A\| = \infty$ for A in (1).

Boundedness and continuity

Theorem

Let E and E' be normed linear spaces and let $A : E \longrightarrow E'$ be a linear transformation. Then the following conditions on A are all *equivalent to one another*:

- (1) A is *continuous*.
- (2) A is *continuous at the origin*, in the sense that $v_n \rightarrow 0 \Rightarrow A(v_n) \rightarrow 0$.
- (3) The *norm $\|A\|$ of A is finite*.
- (4) If $B = \{v : \|v\| \leq 1\}$ is the closed unit ball in E , then its image $A(B)$ is a *bounded set* in E' .

This theorem explains why in the context on linear maps the terms *bounded* and *continuous* are used interchangeably when referring to linear maps. The set of such maps is denoted by $L(E, E')$

The normed space $(L(E, E'), \|\cdot\|)$

Theorem

Let $(E, \|\cdot\|)$ and $(E', \|\cdot\|)$ be two linear normed spaces, then:

- (1) The space $(L(E, E'), \|\cdot\|)$ is a normed space.
- (2) If $(E', \|\cdot\|)$ is a Banach space then so is $(L(E, E'), \|\cdot\|)$.

Outline

- 1 Completeness and Banach spaces
- 2 Fixed Point Theorems**
- 3 Application 1: Newton's Method
- 4 Application 2: Initial-Value Problems and Picard's Method
- 5 References

Fixed Point Theorems

Fixed Point Theorems are central results in mathematics. We will see two applications:

- (i) The **Picard method** for the solution of differential equations:
Find a curve $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{R}^n$ that satisfies

$$\frac{d\mathbf{x}(t)}{dt} = F(t, \mathbf{x}(t)), \text{ with } \mathbf{x}(0) = \mathbf{x}_0 \in \mathbb{R}^n \text{ (initial condition)} \quad (2)$$

for some prescribed function $F : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$.

Can be formulated as a fixed point condition.

$\leadsto$ find a Banach space X of curves $\mathbf{x}(t)$ and consider the map

$$T : X \rightarrow X, \quad T(\mathbf{x})(t) = \mathbf{x}_0 + \int_0^t F(s, \mathbf{x}(s)) ds,$$

Then $T(\mathbf{x}) = \mathbf{x}$ if and only if $\mathbf{x}(t)$ satisfies (2).

(ii) **Newton's method** to find the roots of a nonlinear equation.

Let $f \in C^1(\mathbb{R})$ such that $f' \in C^0(\mathbb{R})$ is never zero. We say that $x_0 \in \mathbb{R}$ is a **root** of f whenever $f(x_0) = 0$.

We define its **Newton function** by $F(x) = x - \frac{f(x)}{f'(x)}$. It is clear that

$$f(x_0) = 0 \iff F(x_0) = x_0. \quad (3)$$

Important implications in **optimization** since extrema are sometimes the roots of the derivatives.

We will address the following questions about fixed points

- When do fixed points **exist**?: **Brouwer's** and **Schauder's** Theorems (no proofs).
- When are fixed points **unique**?: **Banach's Contraction-Mapping Principle** – **crucial use of the concept of completeness!**

A first existence theorem

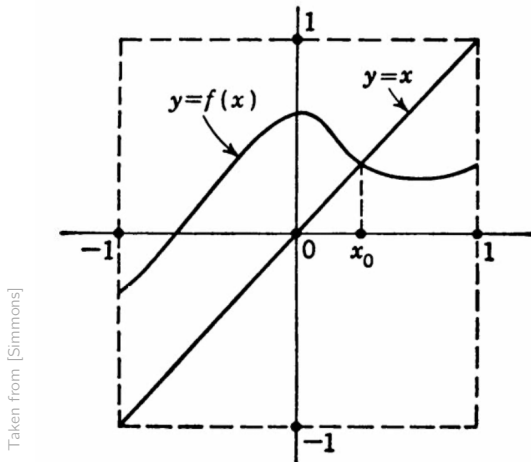
Theorem

Every *continuous map* $f : [-1, 1] \longrightarrow [-1, 1]$ *has a fixed point*.

Proof

Suppose that f does not fix either endpoint (otherwise we are done). This implies that $f(-1) > -1$ and $f(1) < 1$. Consider now the function $g(x) = f(x) - x$. By construction g is continuous, $g(-1) > 0$, and $g(1) < 0$. The Intermediate Value Theorem implies that there exists $x_0 \in (-1, 1)$ such that

$$g(x_0) = 0 \iff f(x_0) = x_0. \quad \blacksquare$$



What about higher dimensions? What is a natural generalization of the interval $[-1, 1]$?

Theorem (Brouwer's Fixed Point Theorem)

Let $B = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq 1\}$ be the *closed unit ball* in $\mathbb{R}^n$. Any continuous map $f : B \rightarrow B$ has a *fixed point*.

- The proof of this result is more difficult and requires the tools of algebraic topology.
- We can generalize this result further even to infinite dimensions: we can go from vectors to functions!

Theorem (Schauder's Fixed Point Theorem)

Let S be a *compact* (later...) and *convex* subset of a *Banach space*. Any continuous map $f : S \rightarrow S$ has a *fixed point*.

What about uniqueness?

Up until now it was the properties of the space that ensured the **existence** of fixed points. **Uniqueness** requires conditions on the function whose fixed points we want to study.

Definition

Let (X, d) be a metric space and let $T : X \rightarrow X$ be a map. We say that T is a **(strict) contraction** when there exists a constant $0 < r < 1$ such that

$$d(T(x), T(y)) < rd(x, y), \quad \text{for all } x, y \in X.$$

Note that contractions are automatically **uniformly continuous** ($\delta = \delta(\epsilon)$ only, i.e., independent of x , in the definition of continuity) and **Lipschitz** ($r \geq 0$).

Examples of contractions

Linear contractive maps

Let $(E, \|\cdot\|)$ be a **normed space** and consider the associated operator norm $\|A\|$ of a linear map $A : E \rightarrow E$ as defined before. Then **linear maps** A with operator norm $\|A\| < 1$ are **contractions**:

$$\|Av - Aw\| = \|A(v - w)\| \leq \|A\| \|v - w\|.$$

Nonlinear contractive maps

Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a map in $C^1(\mathbb{R}^n, \mathbb{R}^n)$ and let $\|\cdot\|$ be a norm in $\mathbb{R}^n$. Let $Df(\mathbf{x}) : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be the **differential map** of f at the point $\mathbf{x} \in \mathbb{R}^n$. This is a **linear map** whose matrix expression using the canonical basis in $\mathbb{R}^n$ is the **Jacobian matrix**

$$Df(\mathbf{x}) = \left(\frac{\partial f_i(\mathbf{x})}{\partial x_j} \right)_{i,j \in \{1, \dots, n\}}.$$

If $\sup_{\mathbf{x} \in \mathbb{R}^n} \|Df(\mathbf{x})\| = r < 1$ for any $\mathbf{x} \in \mathbb{R}^n$, then f is a **contraction** in $\mathbb{R}^n$ with respect to the metric induced by $\|\cdot\|$. Indeed by the **Mean Value Theorem**, for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ there exists $\lambda \in (0, 1)$ such that

$$f(\mathbf{y}) - f(\mathbf{x}) = Df((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) \cdot (\mathbf{y} - \mathbf{x}).$$

$$\begin{aligned} \|f(\mathbf{y}) - f(\mathbf{x})\| &= \|Df((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) \cdot (\mathbf{y} - \mathbf{x})\| \\ &\leq \|Df((1 - \lambda)\mathbf{x} + \lambda\mathbf{y})\| \|\mathbf{y} - \mathbf{x}\| \leq r \|\mathbf{y} - \mathbf{x}\|. \end{aligned}$$

Banach Contraction-Mapping Principle

Theorem

If $T : X \longrightarrow X$ is a *contraction* defined on a *complete metric space* (X, d) , then T *has a unique fixed point*.

Proof

Existence: Let x_0 be an arbitrary point in X , and write $x_1 = Tx_0$, $x_2 = T^2x_0 = Tx_1$, and, in general, $x_n = T^n x_0 = Tx_{n-1}$. If $m < n$, then

$$\begin{aligned} d(x_m, x_n) &= d(T^m x_0, T^n x_0) = d(T^m x_0, T^m T^{n-m} x_0) \\ &\leq r^m d(x_0, T^{n-m} x_0) = r^m d(x_0, x_{n-m}) \\ &\leq r^m [d(x_0, x_1) + d(x_1, x_2) + \cdots + d(x_{n-m-1}, x_{n-m})] \\ &\leq r^m d(x_0, x_1) [1 + r + \cdots + r^{n-m-1}] \\ &= r^m d(x_0, x_1) \frac{1 - r^{m-n}}{1 - r} < \frac{r^m d(x_0, x_1)}{1 - r}. \end{aligned}$$

Proof (continued)

Since $r < 1$, it is evident from this that $\{x_n\}$ is a Cauchy sequence, and by the completeness of X , there exists a point x in X such that $x_n \rightarrow x$. We now use the continuity of T to infer that x is a fixed point:

$$Tx = T\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} Tx_n = \lim_{n \rightarrow \infty} x_{n+1} = x.$$

Uniqueness: If y is also a fixed point, that is, if $Ty = y$, then $d(x, y) = d(Tx, Ty) \leq rd(x, y)$; and since $r < 1$, this implies that $d(x, y) = 0$, i.e., $y = x$. ■

How to find the fixed points?

In the presence of contractions, Banach's Theorem does not only prove the existence of fixed points but **tells us how to find them numerically: just iterate the contraction!**

Proposition

If (X, d) is a metric space, $T : X \rightarrow X$ is continuous, and x_0 is a point of X for which **the iterate sequence** $\{x_0, T(x_0), T(T(x_0)), \dots\}$ **converges**, then **the limit** of that sequence **has to be a fixed point of T** .

Proof

Suppose the iterate sequence $\{x_n\}$ of x_0 converges to $p \in X$, that is, $\lim_{n \rightarrow \infty} d(x_n, p) = 0$. Then the continuity of T insures that $x_{n+1} = T(x_n) \rightarrow T(p)$. Also $\lim_{n \rightarrow \infty} d(x_{n+1}, p) = 0$, that is, $x_{n+1} \rightarrow p$, so (because limits in metric spaces are unique) $p = T(p)$. ■

If we assume that T is a (strict) contraction, then there results a very strong converse: we can obtain the fixed point by iterating from any point!

Proposition

Suppose T is a strict contraction on a metric space. If p is a fixed point of T then every iterate sequence converges to p .

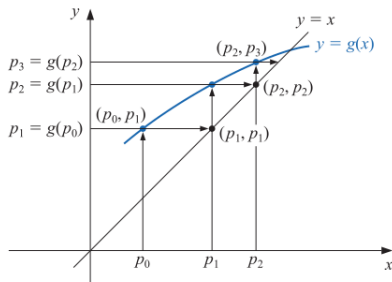
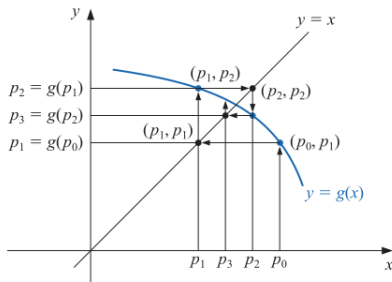
Proof

Let $0 < r < 1$ denote the contraction constant of the mapping T . Fix $x_0 \in X$ and define the iterate sequence $\{x_n\}$ in the usual way:
 $x_1 = T(x_0), \dots, x_n = T(x_{n-1}), \dots$ Then

$$d(x_n, p) = d(T(x_{n-1}), T(p)) \leq r d(x_{n-1}, p) \leq \dots \leq r^n d(x_0, p),$$

so $d(x_n, p) \rightarrow 0$ as $n \rightarrow \infty$, that is, $\{x_n\}$ converges to p . ■

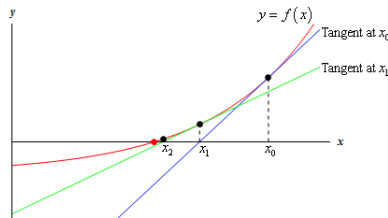
These results produce a **numerical solution** for the fixed point problem for which **no algebraic solution** exists.



Outline

- 1 Completeness and Banach spaces
- 2 Fixed Point Theorems
- 3 Application 1: Newton's Method**
- 4 Application 2: Initial-Value Problems and Picard's Method
- 5 References

First application: Newton's root method



The blue line is given by the function $y = f(x_0) + f'(x_0)(x - x_0)$ and hence the point x_1 is given by $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$, that is, it is the image of the **Newton function** in the introduction (slide 18):

$$F(x) = x - \frac{f(x)}{f'(x)}$$

whose fixed points are the roots of f (for non-zero derivatives).

Setup of Newton's method

Suppose f is a real-valued function defined on a closed interval $I = [a, b]$ and that we know f has a root in the open interval (a, b) . We use Banach's Theorem to show that, under suitable hypotheses on f , Newton's method converges to this root.

More precisely, suppose that $f \in C^2(I)$ with f' never zero on I and that f has different signs at the endpoints of I ; say (without loss of generality) $f(a) < 0$ and $f(b) > 0$. Then f has a unique root x^* in the interior (a, b) of I . Under these hypotheses we have:

Theorem

There exists $\delta > 0$ such that F is a strict contraction on $[x^ - \delta, x^* + \delta]$. As a consequence, for every x_0 in $[x^* - \delta, x^* + \delta]$, Newton's method with starting point x_0 converges to x^**

Notes:

- Newton's method $\equiv$ iterative application of F ;
- the root x^* of f is the fixed point of F .

Said differently, under reasonable hypotheses on f , for starting points close enough to a root of f the iterate sequence for the Newton function

$$F(x) = x - \frac{f(x)}{f'(x)} \quad (x \in I)$$

will converge to that root.

Proof

Let M denote the maximum of $|f''(x)|$ as x ranges through I , and let m denote the corresponding minimum of $|f'(x)|$. By the continuity of f'' , and the hypothesis that f' never vanishes on I , we know that M is finite and $m > 0$. Differentiation of F via the quotient rule yields

$$F'(x) = \frac{f(x)f''(x)}{f'(x)^2} \quad (x \in I)$$

which, along with our bounds on f' and f'' , provides the estimate

$$|F'(x)| \leq \frac{M}{m^2} |f(x)| \quad (x \in I_\delta).$$

Proof (continued I)

Thus, upon shrinking δ enough to insure that

$$|f(x)| \leq \frac{m^2}{2M} \quad \text{for} \quad |x - x^*| < \delta$$

(possible because f is continuous at x^* and takes the value zero there) we see that $|F'(x)| \leq 1/2$ for each $x \in I_\delta = [x^* - \delta, x^* + \delta]$.

Now, if $x, y \in I_\delta$ then, the Mean-Value Theorem it shows that

$$|F(x) - F(y)| = |F'(\bar{x})(x - y)| \leq \frac{1}{2}|x - y| \quad \forall x, y \in I_\delta.$$

Proof (continued II)

Thus F is a strict contraction on I_δ once we know F maps that interval into itself. But it does, since the same inequality shows that for each $x \in I_\delta$ (upon recalling that the root x^* of f is a fixed point of F):

$$|F(x) - x^*| = |F(x) - F(x^*)| \leq \frac{1}{2} |x - x^*| \leq \frac{1}{2} \delta < \delta,$$

so $F(x) \in I_\delta$, as desired.

Thus Banach's Contraction-Mapping Principle applies to the strict contraction F acting on the complete metric space $I_\delta = [x^* - \delta, x^* + \delta]$, and guarantees that for every starting point in I_δ the corresponding F -iteration sequence converges to the fixed point of F , which must necessarily be the unique root of f in I_δ . ■

Outline

- 1 Completeness and Banach spaces
- 2 Fixed Point Theorems
- 3 Application 1: Newton's Method
- 4 Application 2: Initial-Value Problems and Picard's Method**
- 5 References

Picard's Method for the Initial-Value Problem

We now provide a solution to the so-called **initial-value problem** by invoking **Banach's Theorem** for a continuous map on the **Banach space** $(C([a_1, a_2]), \|\cdot\|_\infty)$ of continuous functions defined on the interval $[a_1, a_2]$. This space is a particular case of the space $(\mathcal{C}_B(X, \mathbb{R}), \|\cdot\|_\infty)$ introduced in slide 8, where X is a topological space.

Theorem (Picard's Theorem)

Let $F : R = [a_1, a_2] \times [b_1, b_2] \longrightarrow \mathbb{R}$. If $F(t, x)$ and $\partial F / \partial x(t, x)$ are **continuous** on the closed rectangle R , and if (t_0, x_0) is an interior point of R , then **the differential equation**

$$\frac{dx}{dt} = F(t, x(t))$$

has a unique solution $x(t)$ which passes through (t_0, x_0) .

Strategy: prove that the linear map " $x(t)$ " $\longmapsto$ " $x_0 + \int_{t_0}^t F(s, x(s))ds$ " is a strict contraction.

Proof

Since $F(t, x)$ and $\frac{\partial F}{\partial x}(t, x)$ are continuous in R , they are bounded, and consequently there exist constants K and M such that

$$\begin{aligned} |F(t, x)| &\leq K \\ \left| \frac{\partial F}{\partial x}(t, x) \right| &\leq M \end{aligned} \tag{4}$$

and for all points (t, x) in R .

Step 1: We use the [Mean Value Theorem](#). For any (t, x_1) and (t, x_2) in R the Mean Value Theorem guarantees that

$$|F(t, x_1) - F(t, x_2)| = |x_1 - x_2| \left| \frac{\partial F}{\partial x}(t, x_1 + \theta(x_2 - x_1)) \right|$$

for some θ such that $0 < \theta < 1$. It now follows that

$$|F(t, x_1) - F(t, x_2)| \leq M |x_1 - x_2|, \text{ for all } (t, x_1), (t, x_2) \text{ in } R. \tag{5}$$

Proof (Continued I)

Step 2: The integral equation formulation: we replace our problem by an equivalent problem relating to an integral equation. Indeed, $x(t)$ is a solution of the initial-value problem if and only if

$$x(t) - x(t_0) = \int_{t_0}^t F(s, x(s)) ds,$$

$$x(t) = x_0 + \int_{t_0}^t F(s, x(s)) ds.$$

It hence suffices to show that this integral equation has a unique solution.

Proof (Continued II)

Step 3: The operator: choose first a positive number a such that $Ma < 1$ and such that the closed rectangle $R' = [t_0 - a, t_0 + a] \times [x_0 - Ka, x_0 + Ka]$ is included in R .

Let X be the set of all continuous real functions $x(t)$ defined on the closed interval $|t - t_0| \leq a$ such that $|x(t) - x_0| \leq Ka$.

X is a closed subspace of the Banach space $(C([t_0 - a, t_0 + a]), \|\cdot\|_\infty)$ and is hence a complete metric space.

Define the mapping T of X into itself defined by

$$\begin{aligned} T : X &\longrightarrow X \\ x &\longmapsto T(x)(t) = x_0 + \int_{t_0}^t F(s, x(s)) ds. \end{aligned}$$

T maps X into itself because by (4),

$$|T(x)(t) - x_0| = \left| \int_{t_0}^t F(s, x(s)) ds \right| \leq Ka.$$

Proof (Continued III)

Step 4: T is a contraction: Let $x_1, x_2 \in X$. It follows from (5) that

$$\begin{aligned} |T(x_1)(t) - T(x_2)(t)| &= \left| \int_{t_0}^t [F(s, x_1(s)) - F(s, x_2(s))] ds \right| \\ &\leq Ma \sup_{t \in [t_0-a, t_0+a]} |x_1(t) - x_2(t)|, \end{aligned}$$

which implies that

$$\|T(x_1) - T(x_2)\|_{\infty} \leq Ma \|x_1 - x_2\|_{\infty},$$

and since $Ma < 1$, this shows that T is a contraction on X .

Using Banach's Contraction Principle we can conclude that the equation $T(x) = x$ has a unique solution. Since this amounts to saying that the integral equation has a unique solution, our proof is complete. ■

The ideas in this proof have a much wider scope than might be suspected, and can be applied to establish many other existence theorems in the theory of differential and integral equations.

Outline

- 1 Completeness and Banach spaces
- 2 Fixed Point Theorems
- 3 Application 1: Newton's Method
- 4 Application 2: Initial-Value Problems and Picard's Method
- 5 References

References I



James Munkres. Topology. Pearson, second edition, 2014.



George F. Simmons. Topology and Modern Analysis. McGraw-Hill, 1963.